

AKILESH A

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🐙 GitHub 🏆 LeetCode

PROFILE

Integrated MSc Software Systems student with hands-on experience in data analysis, machine learning, predictive modeling, and database-driven applications. Skilled in Python, SQL, Pandas, NumPy, Scikit-learn, PostgreSQL, and data visualization, with a strong foundation in data structures, DBMS, and statistical analysis. Passionate about transforming complex datasets into actionable insights and scalable data solutions.

EDUCATION

Integrated MSc Software Systems, 3rd Year
Coimbatore Institute of Technology

2024 – 2029
Coimbatore

SKILLS

Data Science & ML

Pandas, NumPy, scikit-learn, Predictive Modeling, Data Visualization

Front End Development

HTML, CSS, JavaScript, React.js, Angular

Database & AI

MySQL, MongoDB, PostgreSQL, Oracle, PL/SQL, Prompt Engineering

Programming Languages

Python, Java, C, C++

Back end Development

Node.js, Express.js

Version Control & Collaboration

Git, GitHub

ACADEMIC PROJECTS

VeloPulseAI – Predictive Maintenance & Real-Time Telematics System

Tech Stack: Python, Machine Learning, Scikit-learn, Pandas, NumPy, Predictive Analytics

- Engineered a predictive maintenance pipeline to process multi-sensor vehicle telemetry and identify anomalous mechanical patterns.
- Developed data preprocessing, feature engineering, and time-series analysis workflows to extract degradation and failure indicators from telemetry data.

RouteVeda – Intelligent Travel Planning Platform

Tech Stack: PostgreSQL, PostGIS, REST APIs, Angular, TypeScript, Node.js

- Architected normalized PostgreSQL schemas with PostGIS to analyze geospatial data and support proximity-based queries.
- Integrated REST APIs to ingest, validate, and process dynamic multi-destination location data for route planning and analysis.

Smart Hospital Bed Management System

Tech Stack: MySQL, JavaScript, PHP, 8086 Assembly Language, HTML, CSS

- Developed a healthcare resource analytics system to track patient intake, bed capacity, oxygen inventory, and physician availability.
- Implemented rule-based triage classification and structured MySQL schemas to analyze admissions, discharges, and ward utilization.

FinvisIQ — Personal Finance Intelligence Platform

Tech Stack: MySQL, Java, JavaFX, CSS

- Developed a personal finance application for expense tracking, budgeting, financial insights, and data visualization.
- Designed modular application workflows for financial record management and analytics, improving maintainability and structured data handling.

PARTICIPATIONS & CERTIFICATIONS

SmartIndia Hackathon 2026 — Internal Round

Contributed to VeloPulseAI, an AI-powered predictive maintenance and vehicle telematics solution, as part of a multidisciplinary team.

Introduction to Front-End Development

Trained in HTML, CSS, JavaScript, responsive web development, and UI engineering.